Hewitt, J. and Manning, C. D. A structural probe for finding syntax in word representations. *arXiv preprint arXiv:1905.06316*, 2019.

Hoerl, A. E. and Kennard, R. W. Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67, 1970.

Hou, A., Zhang, J., He, T., Wang, Y., Chuang, Y.-S., Wang, H., Shen, L., Van Durme, B., Khashabi, D., and Tsvetkov, Y. SemStamp: A semantic watermark with paraphrastic robustness for text generation. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 4067–4082, Mexico City, Mexico, June 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-long.226. URL https://aclanthology.org/2024.naacl-long.226/.

Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. LoRA: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2022.

Hugging Face H4 Team. Helpful instructions dataset. https://huggingface.co/datasets/HuggingFaceH4/helpful-instructions, 2023.

Kandpal, N., Jagielski, M., Tramèr, F., and Carlini, N. Backdoor attacks for in-context learning with language models. *arXiv preprint arXiv:2307.14692*, 2023.

Kirchenbauer, J., Geiping, J., Wen, Y., Katz, J., Miers, I., and Goldstein, T. A watermark for large language models. In *Proceedings of the 40th International Conference on Machine Learning*, pp. 17061–17084. PMLR, 2023.

Liu, Y., Ma, S., Aafer, Y., Lee, W.-C., Zhai, J., Wang, W., and Zhang, X. Trojaning attack on neural networks. In *Network and Distributed System Security Symposium (NDSS)*, 2018.

Manakul, P., Liusie, A., and Gales, M. J. F. Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models. *arXiv preprint arXiv:2303.08896*, 2023. URL https://arxiv.org/abs/2303.08896.

Meier, D., Wahle, J. P., Röttger, P., Ruas, T., and Gipp, B. TrojanStego: Your language model can secretly be a steganographic privacy leaking agent. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 27244–27261, Suzhou, China, November 2025. Association for Computational Linguistics. ISBN 979-8-89176-332-6. doi: 10.18653/v1/2025.emnlp-main.1386. URL https://aclanthology.org/2025.emnlp-main.1386/.

Meng, P., Huang, L., Chen, Z., Yang, W., and Li, D. Linguistic steganography detection based on perplexity. In *Proceedings of the 2008 International Conference on MultiMedia and Information Technology (MMIT 2008)*, pp. 217–220, 2008. ISBN 9780769535562. doi: 10.1109/MMIT.2008.29.

Merity, S., Xiong, C., Bradbury, J., and Socher, R. Pointer sentinel mixture models. *arXiv preprint arXiv:1609.07843*, 2016.

Mistral AI team. Un ministral, des ministraux. https://mistral.ai/news/ministraux, October 2024.

OpenAI. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.

Pearson, K. On lines and planes of closest fit to systems of points in space. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 2(11): 559–572, 1901.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

Rafailov, R., Sharma, A., Mitchell, E., Manning, C. D., Ermon, S., and Finn, C. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://papers.nips.cc/paper_files/paper/2023/hash/a85b405ed65c6477a4fe8302b5e06ce7-Abstract-Confere
html.

Rando, J. and Tramèr, F. Universal jailbreak backdoors from poisoned human feedback. *arXiv preprint arXiv:2311.14455*, 2023.

Reimers, N. and Gurevych, I. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*, 2019. URL https://arxiv.org/abs/1908.10084.

Shuttleworth, R., Andreas, J., Torralba, A., and Sharma, P. Lora vs full fine-tuning: An illusion of equivalence. *arXiv preprint arXiv:2410.21228*, 2024. doi: 10.48550/arXiv.2410.21228.

Simmons, G. J. The prisoners' problem and the subliminal channel. In *Advances in Cryptology*. Springer, 1984.

Taori, R., Gulrajani, I., Zhang, T., Dubois, Y., Li, X., Guestrin, C., Liang, P., and Hashimoto, T. B. Stanford alpaca: An instruction-following llama model. https://crfm.stanford.edu/2023/03/13/alpaca.html, 2023.

Tenney, I., Das, D., and Pavlick, E. BERT rediscovers the classical NLP pipeline. *arXiv preprint arXiv:1905.05950*, 2019.

Tong, Y. et al. Seedprints: Fingerprints can even tell which seed your large language model was trained from. *arXiv preprint arXiv:2509.26404*, 2025.

Tramèr, F., Zhang, F., Juels, A., Reiter, M. K., and Ristenpart, T. Stealing machine learning models via prediction apis. In *USENIX Security Symposium*, 2016.

Wang, Y., Tang, Y., et al. State-of-the-art advances of deep-learning linguistic steganalysis research. *arXiv preprint arXiv:2409.01780*, 2024.

Yan, R. and Murawaki, Y. Addressing tokenization inconsistency in steganography and watermarking based on large language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 7076–7098, Suzhou, China, November 2025. Association for Computational Linguistics. ISBN 979-8-89176-332-6. doi: 10.18653/v1/2025.emnlp-main.361. URL https://aclanthology.org/2025.emnlp-main.361/.

Yang, Z., Wei, N., Liu, Q., Huang, Y., and Zhang, Y. Gantstega: Text steganography based on generative adversarial networks. In *Digital Forensics and Watermarking (IWDW 2019)*, volume 12022 of *Lecture Notes in Computer Science*, pp. 18–31. Springer, Cham, 2020. doi: 10.1007/978-3-030-43575-2_2.

Youstra, J., Mahfoud, M., Yan, Y., Sleight, H., Perez, E., and Sharma, M. Towards safeguarding LLM fine-tuning APIs against cipher attacks, August 2025. URL https://arxiv.org/abs/2508.17158.

Ziegler, Z., Deng, Y., and Rush, A. Neural linguistic steganography. *arXiv preprint arXiv:1909.01496*, 2019.

A. Notation

Table 4. Notation used throughout the paper.

Symbol	Meaning
$\mathcal{V}$	Vocabulary
$V := \lvert\mathcal{V}\rvert$	Vocabulary size
$t \in \mathcal{V}$	Token
$\mathbf{E} \in \mathbb{R}^{V\times d}$	Token embedding matrix
$\mathbf{e}_t \in \mathbb{R}^d$	Embedding of token t (row of $\mathbf{E}$)
d	Embedding dimension
σ	Seed defining a random hyperplane
$\mathbf{v}_\sigma \in \mathbb{R}^d$	Unit normal vector for seed σ
τ	Hyperplane threshold (e.g., median projection)
$\beta_\sigma(t) \in \{0,1\}$	Bucket assignment of token t under seed σ
S	Secret (4-letter string)
$\mathrm{Encode}(\cdot)$	Secret-to-bits encoding function, $\mathrm{Encode}(S) \in \{0,1\}^{L_{\text{secret}}}$
L_{secret}	Bit-length of $\mathrm{Encode}(S)$ (in our experiments, $L_{\text{secret}} = 32$)
L_{tx}	Number of payload positions / transmitted bits (e.g., 32 in Wiki setup; 128 if the 32-bit code is repeated 4 times)
$\mathbf{b} = (b_1,\ldots,b_{L_{\text{tx}}}) \in \{0,1\}^{L_{\text{tx}}}$	Transmitted bitstring (payload) across payload positions
$O = (O_1,\ldots,O_{L_{\text{tx}}})$	Output token sequence at payload positions
$\mathcal{H}$	Classifier family for bit recovery from O
$\mathcal{G}$	Classifier family for secret recovery from $\mathbf{b}$
$\mathcal{R}_{\text{bucket}}$	Bit recoverability: $\max_{h\in\mathcal{H}} \mathrm{Acc}(h(O),\mathbf{b})$
$\mathcal{R}_{\text{secret}}$	Secret recoverability: $\max_{h\in\mathcal{H},\, g\in\mathcal{G}} \mathrm{Acc}(g(h(O)),S)$
c	Decoding context (prefix) in next-token distributions
$P(\cdot \mid c)$	Base next-token distribution
$Q(\cdot \mid c)$	Steganographic (constrained) next-token distribution
Q^*	Optimal distribution under KLD constraint
$\mathcal{Q}$	Set of allowed steganographic (constrained) next-token distributions
n	Number of buckets ($\log_2 n$ bits per token)
$t^\star$	Token promoted to argmax in KL/PPL bounds

B. Proofs

B.1. Proof of Theorem 7.1 (KL Lower Bound)

Proof. The proof proceeds in two parts: first we derive the pooled-and-rescaled form of the optimal Q^* via KKT conditions, then we substitute this form into the KL divergence and show it collapses to $\log\alpha$.

Part I: KKT derivation of the pooling structure. We solve the constrained optimization problem

$$\min_{q\in\mathbb{R}^V} \sum_{t=1}^{V} q_t \log\frac{q_t}{p_t} \quad \text{s.t.} \quad \sum_{t=1}^{V} q_t = 1, \quad q_t - q_{t^\star} \le 0 \;\; \forall\, t \ne t^\star.$$

The Lagrangian with multiplier λ for normalization and $\mu_t \ge 0$ for the inequality constraints is

$$\mathcal{L}(q,\lambda,\mu) = \sum_{t=1}^{V} q_t \log\frac{q_t}{p_t} + \lambda\Big(\sum_{t=1}^{V} q_t - 1\Big) + \sum_{t\ne t^\star} \mu_t (q_t - q_{t^\star}).$$

Stationarity $\partial\mathcal{L}/\partial q_t = 0$ gives, for $t \ne t^\star$:

$$\log\frac{q_t}{p_t} + 1 + \lambda + \mu_t = 0,$$

and for $\partial\mathcal{L}/\partial q_t^* = 0$ implies:

$$\log\frac{q_{t^\star}}{p_{t^\star}} + 1 + \lambda - \sum_{t\ne t^\star} \mu_t = 0.$$

Complementary slackness $\mu_t(q_t - q_{t^\star}) = 0$ naturally partitions $\mathcal{V}\setminus\{t^\star\}$ into two sets:

$$\mathcal{U} := \{t \ne t^\star : q_t < q_{t^\star}\}, \qquad \mathcal{C} := \{t \ne t^\star : q_t = q_{t^\star}\}, \qquad k := |\mathcal{C}|.$$